ABSTRACT OF THE DISCLOSURE

An apparatus for a computer data bus with a data path width greater than 128 bits, implemented via a plurality of parallel optical channels. The apparatus overcomes the physical limitations of electrical buses, such as signal crosstalk, timing skew, and pin density, which have made architectures wider than 64 bits impractical. By grouping a small number of high-cardinality optical channels to operate in concert, the invention enables the creation of ultra-wide data buses in a microscopic footprint, providing a foundational component for next-generation, high-throughput computing architectures.